

# THE JACOBI INVERSION PROBLEM AND THETA DIVISORS

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ABSTRACT. An expository article on solving the Jacobi Inversion problem for compact Riemann surfaces of genus  $g \geq 2$  using Theta divisors, following Chapters 2.4-2.7 of Principles of Algebraic Geometry by Philip Griffiths and Joe Harris.

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## 1. INTRODUCING THE JACOBI INVERSION PROBLEM

The study of compact Riemann surfaces goes back to the study of integrals of the form

$$\int \frac{dx}{\sqrt{x^3 + ax^2 + bx + c}}.$$

Mathematicians tried to study these but soon ran into an issue: these integrals are not well-defined. One way to see this is to view this integral as a line integral

$$\int \frac{dx}{y}$$

of the everywhere holomorphic differential  $\frac{dx}{y}$  on the cubic curve  $C$  defined by  $y^2 = x^3 + ax^2 + bx + c$ . When  $x^3 + ax^2 + bx + c$  has distinct roots,  $C$  is a smooth curve of genus 1, and topologically a torus. Thus, such an integral is only well defined modulo the periods  $\int_{\gamma} \frac{dx}{y}$  over closed loops  $\gamma \in H_1(C, \mathbb{Z})$ . While such integrals are only well-defined modulo periods, in 1826 Abel noticed that sums of these integrals follow certain qualitative behaviors. For example, any line  $L \subset \mathbb{P}^2$  will intersect  $C$  at points  $p_1(L), p_2(L), p_3(L)$ , not necessarily distinct and of no particular order. Fixing a base point  $b$ , and letting  $\Lambda \subset \mathbb{C}$  denote the

period lattice, Abel found that the sum

$$\left[ \int_b^{p_1(L)} \omega + \int_b^{p_2(L)} \omega + \int_b^{p_3(L)} \omega \right] \mod \Lambda.$$

is constant modulo the period lattice. This example is a special instance of injectivity of the Abel-Jacobi map, which we now define in generality.

Let us first mention the general setup. A good reference for all of this is the first three sections of Chapter 2 of [GH94]. Let  $S$  be a compact Riemann surface of genus  $g$ . Let  $\gamma_1, \dots, \gamma_g, \gamma_{g+1}, \dots, \gamma_{2g} \in H_1(S, \mathbb{Z})$  be a canonical basis of loop generators. Let  $\omega_1, \dots, \omega_g$  be a basis for  $H^0(S, \Omega_S)$ . Then the period lattice  $\Lambda$  is generated by integral linear combinations of  $\lambda_1, \dots, \lambda_{2n}$ , where

$$\lambda_j = \left( \int_{\gamma_j} \omega_1 \quad \cdots \quad \int_{\gamma_j} \omega_g \right).$$

The Jacobian  $\mathcal{J}(S)$  of  $S$  is defined to be the  $g$ -dimensional torus  $\frac{\mathbb{C}^g}{\Lambda}$ . Choosing  $e_1, \dots, e_g$  to be the standard complex basis of  $\mathbb{C}^g$ , the period matrix of  $\mathcal{J}(S)$  is simply the  $g \times 2g$  matrix

$$\begin{pmatrix} \lambda_1^T & \cdots & \lambda_{2n}^T \end{pmatrix}.$$

Choosing a normalized basis for  $H^0(S, \Omega_S)$  induces a basis  $\lambda_1, \dots, \lambda_{2n}$  such that the period matrix is of the form  $\begin{pmatrix} I & Z \end{pmatrix}$  where  $Z$  is symmetric and  $\Im(Z)$  is positive definite, which demonstrates that the Jacobian  $\mathcal{J}(S)$  of a compact Riemann surface  $S$  of genus  $g$  is principally polarized.

The Abel-Jacobi map of  $S$  is defined to be the map

$$\text{Pic}^0(S) \rightarrow \mathcal{J}(S)$$

where for any degree 0 divisor  $\sum(p_i - q_i)$ ,

$$\mu\left(\sum(p_i - q_i)\right) = \left( \sum \int_{q_i}^{p_i} \omega_1, \dots, \sum \int_{q_i}^{p_i} \omega_g \right).$$

A celebrated fact is the following.

**Theorem 1.1.** [GH94, Page 235] *The Abel-Jacobi map is an isomorphism.*

The geometric interpretation of the injectivity of the Abel-Jacobi map is straightforward. Given a collection of paths  $\{\tau_1, \dots, \tau_s | \tau_i : [0, 1] \rightarrow S\}$  on  $S$  and another collection of paths  $\{\tau'_1, \dots, \tau'_s | \tau'_i : [0, 1] \rightarrow S\}$  on  $S$ , then integrating along these paths

$$\left( \sum \int_{\tau_i} \omega_1, \dots, \sum \int_{\tau_i} \omega_g \right) = \left( \sum \int_{\tau'_i} \omega_1, \dots, \sum \int_{\tau'_i} \omega_g \right) \mod \Lambda$$

yields the same value in  $\mathcal{J}(S)$  if and only if there exists a meromorphic differential  $\eta$  such that  $\sum(\tau_i(1) - \tau_i(0)) = (\eta) + \sum(\tau'_i(1) - \tau'_i(0))$ .

Surjectivity of the Abel-Jacobi map is typically established by showing that the composite

$$S^{(g)} \rightarrow \text{Pic}^0(S) \rightarrow \mathcal{J}(S)$$

is surjective, where  $S^{(g)}$  is a complex manifold, obtained by quotienting  $S^g$  by the natural action of the symmetric group  $\Sigma_g$  on  $g$  letters, whose points correspond to effective divisors of  $S$  of degree  $g$ . Fixing a base point  $b \in S$ , the composite is then the map

$$\sum_{i=1}^g p_i \mapsto \left( \sum_{i=1}^g \int_b^{p_i} \omega_1, \dots, \sum_{i=1}^g \int_b^{p_i} \omega_g \right).$$

For general  $c \in \Lambda$ , the points  $p_1, \dots, p_g$  are unique. This is because each fiber of the composite is a projective space, since elements belonging to the same fiber are exactly linearly equivalent effective divisors of degree  $g$ , and the general fiber has dimension 0. The Jacobi Inversion problem asks the following question.

**Question 1.2.** *For general  $c \in \mathcal{J}(S)$ , can we explicitly find the  $g$  points  $p_1, \dots, p_g$ , such that*

$$\left( \sum_{i=1}^g \int_b^{p_i} \omega_1, \dots, \sum_{i=1}^g \int_b^{p_i} \omega_g \right) = c \pmod{\Lambda}?$$

In the case of elliptic curves, or compact Riemann surfaces of genus 1, the Jacobi Inversion problem is solved using the classical theory of Weierstrass  $p$ -functions and the theory of doubly periodic meromorphic functions on the complex plane. Many great mathematicians worked on the Jacobi Inversion problem for compact Riemann surfaces of genus  $g \geq 2$ , which spurred the development of algebraic geometry.

In this expository note, we introduce the solution to the Jacobi Inversion problem for genus  $g \geq 2$ . If we let  $\mu$  denote the precomposition of the Abel-Jacobi map with the map  $S \rightarrow \text{Pic}^0(S)$ , where  $p \mapsto p - b$ , then the solution will involve intersecting the curve  $\mu(S) \subset \mathcal{J}(S)$  with a certain effective divisor  $\Theta_c$  of  $\mathcal{J}(S)$  called a Theta divisor. These Theta divisors  $\Theta_c$  will naturally arise as effective divisors associated to certain global sections of certain line bundles on the Jacobian  $\mathcal{J}(S)$ . Thus, to set up our solution, we need to first discuss line bundles on  $\mathcal{J}(S)$ , and more generally, line bundles on compact complex tori.

## 2. (POSITIVE) LINE BUNDLES ON COMPACT COMPLEX TORI

Let  $M = V/\Lambda$  be a complex torus of dimension  $n$ . Let  $\mathcal{L}$  be a line bundle on  $M$ . We will first discuss how every line bundle on  $M$  can be realized as a quotient of the trivial line bundle  $V \times \mathbb{C}$  on  $V$ .

First, note that every line bundle on  $V$  is trivial. Taking sheaf cohomology on the exponential exact sequence on  $V$ , we obtain

$$\cdots \rightarrow H^1(V, \mathcal{O}_V) \rightarrow H^1(V, \mathcal{O}_V^\times) \rightarrow H^2(V, \mathbb{Z}) \rightarrow \cdots,$$

and since  $V \cong \mathbb{C}^n$ , we have that  $H^1(V, \mathcal{O}_V) \cong H^{0,1}(V) = 0$  and  $H^2(V, \mathbb{Z}) = 0$ . Thus,  $H^1(V, \mathcal{O}_V^\times) = 0$ , so every line bundle on  $V$  is trivial. In particular, the pullback line bundle can be globally trivialized:

$$\psi : \pi^* \mathcal{L} \rightarrow V \times \mathbb{C}.$$

If  $\lambda_1, \dots, \lambda_{2n}$  are integral generators for  $\Lambda$ , then note for every  $z$ , the fibers  $\pi^*\mathcal{L}|_z$  and  $\pi^*\mathcal{L}|_{z+\lambda_i}$  are identified with  $\mathcal{L}|_{\pi(z)}$ . Using the trivialization  $\psi$ , we have an identification

$$\mathbb{C} \cong^\psi \pi^*\mathcal{L}|_z = \mathcal{L}|_{\pi(z)} = \pi^*\mathcal{L}|_{z+\lambda_i} \cong^\psi \mathbb{C}$$

given by multiplication by some scalar  $e_{\lambda_i}(z)$ . These  $e_{\lambda_i}(z)$ , which we call multipliers, are everywhere nonzero entire functions. Note that these multipliers satisfy the following relation: for all  $\lambda, \lambda' \in \Lambda$ , we have

$$e_{\lambda+\lambda'}(z) = e_\lambda(z) \cdot e_{\lambda'}(z + \lambda) = e_{\lambda'}(z) \cdot e_\lambda(z + \lambda').$$

Furthermore, note that these multipliers are not uniquely associated to  $\mathcal{L}$ . Given some everywhere nonzero entire function  $m(z)$ , we can simply change the global trivialization  $\psi$  by  $(Id, m) \circ \psi$ . This changes the multipliers to be of the form  $\{e_\lambda(z) \cdot \frac{m(z+\lambda)}{m(z)}\}$ . Any line bundle on  $M$  isomorphic to  $\mathcal{L}$  will also produce the same multipliers up to this kind of multiplication.

We can also go in the reverse direction. Given everywhere nonzero entire functions  $\{e_\lambda(z)\}_{\lambda \in \Lambda}$  satisfying the aforementioned relation, we can define a line bundle  $\mathcal{L}$  on  $M$  by imposing an equivalence relation on  $V \times \mathbb{C}$ , such that

$$(v, c) \sim (v + \lambda, c') \iff c' = e_\lambda(z) \cdot c.$$

A global trivialization of  $\pi^*\mathcal{L}$  will produce the same multipliers  $\{e_\lambda(z) \cdot \frac{m(z+\lambda)}{m(z)}\}$  up to a everywhere nonzero entire function  $m(z)$ . Thus we have a one to one correspondence

$$\frac{\{\text{multipliers } \{e_\lambda(z)\}_{\lambda \in \Lambda}\}}{\sim} \longleftrightarrow \text{Pic}(M)$$

where  $\{e_\lambda(z)\} \sim \{e'_\lambda(z)\} \iff \exists m(z) \in \mathcal{O}_V^\times(V)$  such that  $m(z+\lambda)e_\lambda(z) = e'_\lambda(z)m(z), \forall \lambda \in \Lambda$

Now suppose that  $M$  is a projective torus. In other words, by the Kodaira embedding theorem [Huy05, Proposition 5.3.1], there exists a positive line bundle on  $M$ . In this next part, we demonstrate two things. Firstly, that positive line bundles are classified by their first Chern class, up to translation. This means that if  $\mathcal{L}, \mathcal{L}'$  are positive line bundles such that  $c_1(\mathcal{L}) = c_1(\mathcal{L}')$ , then there exists  $\mu \in M$  such that  $\mathcal{L} = \tau^*\mathcal{L}'$ . Secondly, that positive line bundles, up to translation, admit a relatively simple description in terms of multipliers.

Fix a positive line bundle  $\mathcal{L}$  on  $M$ . Then there exists an integral basis  $\lambda_1, \dots, \lambda_{2n}$  for  $\Lambda$  and complex basis  $e_1, \dots, e_n$  such that  $c_1(\mathcal{L}) \in H^2(M; \mathbb{Z})$  can be written as

$$c_1(\mathcal{L}) = \sum_{\alpha=1}^n \delta_\alpha dx_\alpha \wedge dx_{\alpha+n},$$

where  $x_j$  are coordinates of  $V$  dual to  $\lambda_j$  ([GH94, Page 305]). Note that the projection  $\pi : V \rightarrow M$  factors as

$$V \rightarrow \frac{V}{\mathbb{Z}\langle \lambda_1, \dots, \lambda_n \rangle} \xrightarrow{\pi'} M.$$

Note that  $\frac{V}{\mathbb{Z}\langle\lambda_1, \dots, \lambda_n\rangle} \cong (\mathbb{C}^\times)^n$ . Taking the long exact sequence in sheaf cohomology, we find the terms

$$\cdots \rightarrow H^1((\mathbb{C}^\times)^n, \mathcal{O}_{(\mathbb{C}^\times)^n}) \rightarrow H^1((\mathbb{C}^\times)^n, \mathcal{O}_{(\mathbb{C}^\times)^n}^\times) \rightarrow H^2((\mathbb{C}^\times)^n, \mathbb{Z}) \rightarrow H^2((\mathbb{C}^\times)^n, \mathcal{O}_{(\mathbb{C}^\times)^n}) \rightarrow \cdots$$

where  $H^k((\mathbb{C}^\times)^n, \mathcal{O}_{(\mathbb{C}^\times)^n}) \cong H^{0,k}((\mathbb{C}^\times)^n) = 0$  for  $k \geq 1$  ([GH94, Page 27]), so that the first Chern class map  $c_1 : H^1((\mathbb{C}^\times)^n, \mathcal{O}_{(\mathbb{C}^\times)^n}^\times) \rightarrow H^2((\mathbb{C}^\times)^n, \mathbb{Z})$  is an isomorphism. Thus, we see that line bundles on  $\frac{V}{\mathbb{Z}\langle\lambda_1, \dots, \lambda_n\rangle}$  are classified by their first Chern class. Now note that  $c_1((\pi')^*\mathcal{L}) = (\pi')^*c_1(\mathcal{L})$ , and  $x_{\alpha+n}$  is a well-defined function on  $\frac{V}{\mathbb{Z}\langle\lambda_1, \dots, \lambda_n\rangle}$ , and since

$$\int_{\lambda_i} dx_j = \delta_{ij},$$

we find  $[dx_{\alpha+n}] = 0 \in H^1(\frac{V}{\mathbb{Z}\langle\lambda_1, \dots, \lambda_n\rangle}; \mathbb{Z})$ . Thus,

$$c_1((\pi')^*\mathcal{L}) = 0.$$

This implies  $(\pi')^*\mathcal{L}$  is isomorphic to the trivial line bundle on  $\frac{V}{\mathbb{Z}\langle\lambda_1, \dots, \lambda_n\rangle}$ . Taking a global trivialization on  $\frac{V}{\mathbb{Z}\langle\lambda_1, \dots, \lambda_n\rangle}$  and extending it to a global trivialization for  $\pi^*\mathcal{L}$  on  $V$ , the multipliers produced are such that  $e_{\lambda_i} \equiv 1$  for  $1 \leq i \leq n$ . So given a positive line bundle on  $M$ , we can choose an integral basis for the lattice  $\Lambda$  so that its multipliers  $e_{\lambda_i} \equiv 1$ . In fact, we claim that there is a positive line bundle on  $M$  with a very simple description of its multipliers which has the same first Chern class.

Fix  $\omega$  to be some integral invariant positive definite (1,1) form on  $M$  (by the Lefschetz (1,1) theorem [Huy05, Proposition 3.3.2] we know  $\omega$  comes from the first Chern class of some positive line bundle). Again, we can choose an integral basis  $\lambda_1, \dots, \lambda_{2n}$  of  $\Lambda$  and complex basis  $e_1, \dots, e_n$  such that

$$\omega = \sum_{\alpha=1}^n \delta_\alpha dx_\alpha \wedge dx_{\alpha+n},$$

where  $x_j$  are coordinates on  $V$  dual to  $\lambda_j$ , and  $z_j$  are complex coordinates on  $V$  dual to  $e_j$ . We have the following theorem.

**Theorem 2.1.** [GH94, Page 310] *The line bundle  $\mathcal{L}$  on  $M$  given by the multipliers*

$$e_{\lambda_\alpha}(z) \equiv 1 \text{ and } e_{\lambda_{\alpha+n}}(z) = e^{-2\pi i z_\alpha}, 1 \leq \alpha \leq n$$

*has first Chern class*

$$c_1(\mathcal{L}) = [\omega] \in H^2(M, \mathbb{Z}).$$

*Proof.* It is easy to verify that these functions satisfy the multiplier relation, so that they produce a legitimate line bundle  $\mathcal{L}$  on  $M$ . To show that  $c_1(\mathcal{L}) = [\omega]$ , one needs to first define a hermitian metric  $h$  on  $\mathcal{L}$ . Then one calculates the curvature of the Chern connection of  $(\mathcal{L}, h)$ . If on a local trivialization  $\psi : \mathcal{L} \rightarrow U \times \mathbb{C}$  we let  $h$  denote function on  $U$  where  $h(z) = h(\psi^{-1}e(z), \psi^{-1}e(z))$  where  $e$  is the unit frame for  $U \times \mathbb{C}$ , then the curvature of the Chern connection is locally  $\bar{\partial}\partial \log h$  ([Huy05, Example 4.3.9(iii)]). Thus, the hardest

part of this proof is constructing the hermitian metric  $h$  on  $\mathcal{L}$ . We defer the reader to page 310 of Griffiths and Harris for the metric construction and Chern class verification.  $\square$

Let  $\mathcal{L}$  be the line bundle constructed in theorem 2.1. Let  $\tau_u : M \rightarrow M$  denote the translation automorphism by  $u \in M$ . What are the multipliers  $\{e'_\lambda(z)\}_{\lambda \in \Lambda}$  of  $\tau_u^* \mathcal{L}$ ? We have that

$$e'_{\lambda_\alpha}(z) = e_\lambda(z+u) \equiv 1 \text{ and } e'_{\lambda_{\alpha+n}}(z) = e_{\lambda_{\alpha+n}}(z+u) = e^{-2\pi i z_\alpha(z+u)} = e^{-2\pi i z_\alpha(u)} \cdot e_{\lambda_{\alpha+n}}(z).$$

So the multipliers of  $\tau_u^* \mathcal{L}$  have the same simple form up to some constants. Now we show that if  $\mathcal{L}'$  is another positive line bundle such that  $c_1(\mathcal{L}') = [\omega]$  as well, then there exists  $u \in M$  such that  $\tau_u^* \mathcal{L}' \cong \mathcal{L}$ . Consider the inclusion of exact sequences of sheaves

$$\begin{array}{ccccccc} 0 & \longrightarrow & \underline{\mathbb{Z}}_M & \longrightarrow & \mathcal{O}_M & \longrightarrow & \mathcal{O}_M^\times \longrightarrow 0 \\ & & \parallel & & \uparrow & & \uparrow \\ 0 & \longrightarrow & \underline{\mathbb{Z}}_M & \longrightarrow & \underline{\mathbb{C}}_M & \longrightarrow & \underline{\mathbb{C}}_M^\times \longrightarrow 0 \end{array}$$

This induces a sheaf cohomology diagram

$$\begin{array}{ccccc} H^1(M, \mathcal{O}_M) & \longrightarrow & H^1(M, \mathcal{O}_M^\times) & \xrightarrow{c_1} & H^2(M, \underline{\mathbb{Z}}_M) \\ \pi_1 \uparrow & & \uparrow & & \parallel \\ H^1(M, \underline{\mathbb{C}}_M) & \longrightarrow & H^1(M, \underline{\mathbb{C}}_M^\times) & \longrightarrow & H^2(M, \underline{\mathbb{Z}}_M) \end{array}$$

and we have  $H^1(M, \mathcal{O}_M) \cong H^{0,1}(M)$  by Dolbeaut's theorem [Huy05, Corollary 2.6.21], and  $H^1(M, \underline{\mathbb{C}}_M) \cong H^1(M, \mathbb{C}) \cong H^{1,0}(M) \oplus H^{0,1}(M)$  by the Hodge decomposition [Huy05, Corollary 3.2.12] since  $M$  is compact Kahler. With respect to these identifications, the map  $\pi_1$  is in fact given by projection, so  $\pi_1$  is surjective. Then if  $c_1(\mathcal{L}') = 0$ , it is in the image of  $H^1(M, \mathcal{O}_M) \rightarrow H^1(M, \mathcal{O}_M^\times)$ , but surjectivity of  $\pi_1$  implies that  $\mathcal{L}$  comes from an element of  $H^1(M, \underline{\mathbb{C}}_M^\times)$ , which means that  $\mathcal{L} \in H^1(M, \mathcal{O}_M^\times) \cong \check{H}^1(U, \mathcal{O}_M^\times)$  can be identified with constant Cech cocycles, and thus has constant transition functions. This implies that the pullback of  $\mathcal{L}'$  onto  $V$  can be trivialized to produce constant multipliers.

Now consider  $\mathcal{L}^{-1} \otimes \mathcal{L}'$  on  $M$ . Then  $c_1(\mathcal{L}^{-1} \otimes \mathcal{L}') = 0$ . Then  $\mathcal{L}^{-1} \otimes \mathcal{L}'$  has constant transition functions, and since  $(\pi')^*(\mathcal{L}^{-1} \otimes \mathcal{L}')$  is trivial, this implies that the multipliers for  $\mathcal{L}^{-1} \otimes \mathcal{L}'$  are such that  $e''_{\lambda_\alpha} \equiv 1$  and  $e''_{\lambda_{\alpha+n}} \equiv c_{\lambda_{\alpha+n}}$  are constant for  $1 \leq \alpha \leq n$ . This implies that  $\mathcal{L}'$  has multipliers

$$e'_{\lambda_\alpha}(z) = e_{\lambda_\alpha} \equiv 1, \text{ and } e'_{\lambda_{\alpha+n}}(z) = c_{\lambda_{\alpha+n}} \cdot e_{\lambda_{\alpha+n}}(z)$$

for  $1 \leq \alpha \leq n$ . Then if we define

$$u = \frac{-1}{2\pi i} \sum_{\alpha=1}^n \log c_{\lambda_{\alpha+n}} \cdot e_\alpha,$$

we find  $\mathcal{L}' \cong \tau_u^* \mathcal{L}$ , as desired.

## 3. GLOBAL SECTIONS AND THETA DIVISORS

In this section we compute the dimension of the space of global sections of a positive line bundle on  $M$ . Afterwards, we discuss positive line bundles on the Jacobian  $\mathcal{J}(S)$  of a compact Riemann surface  $S$  of genus  $g$ , and define Theta divisors.

Fix a positive line bundle  $\mathcal{L}$  on  $M = V/\Lambda$ . Choose an integral basis  $\lambda_1, \dots, \lambda_{2n}$  for  $\Lambda$  such that

$$c_1(\mathcal{L}) = \sum_{\alpha=1}^n \delta_\alpha dx_\alpha \wedge dx_{\alpha+n},$$

and let  $\{e_\alpha\}_{\alpha=1}^n$  be a complex basis for  $V$  where  $e_\alpha = \frac{1}{\delta_\alpha} \lambda_\alpha$ . The period matrix for  $M$  is then  $(\Lambda_\delta \quad Z)$  where  $Z$  is symmetric and  $\text{Im}(Z)$  is positive definite. Note that since  $c_1(\mathcal{L})$  is a Kahler form,  $c_1(\mathcal{L})^n \neq 0$ , so the  $\delta_\alpha$  are all nonzero positive integers. We will now prove the following proposition.

**Proposition 3.1.** *The dimension of the space of global sections of  $\mathcal{L}$  is*

$$\dim H^0(M, \mathcal{L}) = \prod_{\alpha=1}^n \delta_\alpha.$$

*Proof.* Note that it suffices to prove this proposition for any line bundle with equal Chern class. This is because any such line bundle is a positive line bundle which, by the work we did in the previous section, differs from  $\mathcal{L}$  by only a translation by some  $u \in M$ . Pulling back  $\mathcal{L}$  by translation automorphism will neither affect the elementary divisors, as  $\sum_{\alpha=1}^n \delta_\alpha dx_\alpha \wedge dx_{\alpha+n}$  is a translation-invariant 2-form, nor affect the dimension of the space of global sections.

Then consider the line bundle given by multipliers

$$e_{\lambda_\alpha} \equiv 1 \text{ and } e_{\lambda_{\alpha+n}} = e^{-2\pi i z_\alpha}, 1 \leq \alpha \leq n,$$

and pull back this line bundle by  $\tau_u$  where  $u = \frac{1}{2} \sum Z_{\alpha\alpha} \cdot e_\alpha$ . Then the multipliers for this new line bundle, which we call  $\mathcal{L}$ , has multipliers

$$e_{\lambda_\alpha} \equiv 1 \text{ and } e_{\lambda_{\alpha+n}} = e^{-2\pi i z_\alpha - \pi i Z_{\alpha\alpha}}.$$

Then we see that global sections of  $\mathcal{L}$  are given by entire functions  $\theta$  which satisfy

$$\theta(z + \lambda_\alpha) = e_{\lambda_\alpha}(z) \cdot \theta(z) = \theta(z) \text{ and } \theta(z + \lambda_{\alpha+n}) = e_{\lambda_{\alpha+n}}(z) \cdot \theta(z) = e^{-2\pi i z_\alpha - \pi i Z_{\alpha\alpha}} \cdot \theta(z).$$

Let  $z_\alpha^* = e^{2\pi i \frac{1}{\lambda_\alpha} z_\alpha}$ . The condition that  $\theta(z + \lambda_\alpha) = \theta(z)$  implies that we can write

$$\theta = \sum_{\ell \in \mathbb{Z}^n} a_\ell (z_1^*)^{\ell_1} \cdots (z_n^*)^{\ell_n} = \sum_{\ell \in \mathbb{Z}^n} a_\ell e^{2\pi i \langle \ell, \Lambda_\delta^{-1} z \rangle}.$$

Now we unpack what the condition  $\theta(z + \lambda_{\alpha+n}) = e^{-2\pi i z_\alpha - \pi i Z_{\alpha\alpha}} \cdot \theta(z)$  tells us about this expression of  $\theta$ .

We have that

$$\theta(z + \lambda_{\alpha+n}) = \sum_{\ell \in \mathbb{Z}^n} a_\ell e^{2\pi i \langle \ell, \Delta_\delta^{-1}(z + \lambda_{\alpha+n}) \rangle} = \sum_{\ell \in \mathbb{Z}^n} a_\ell e^{2\pi i \langle \ell, \Delta_\delta^{-1} \lambda_{\alpha+n} \rangle} \cdot e^{2\pi i \langle \ell, \Delta_\delta^{-1} z \rangle},$$

and the condition implies that this is equal to

$$= e^{-2\pi i z_\alpha - \pi i Z_{\alpha\alpha}} \cdot \sum_{\ell \in \mathbb{Z}^n} a_\ell e^{2\pi i \langle \ell, \Delta_\delta^{-1} z \rangle} = e^{-\pi i Z_{\alpha\alpha}} \sum_{\ell \in \mathbb{Z}^n} a_{\ell + \delta_\alpha \cdot e_\alpha} e^{2\pi i \langle \ell, \Delta_\delta^{-1} z \rangle},$$

since  $e^{-2\pi i z_\alpha} = (z_\alpha^*)^{-\delta_\alpha}$ . Altogether, this implies that

$$a_{\ell + \delta_\alpha \cdot e_\alpha} = e^{2\pi i \langle \ell, \Delta_\delta^{-1} \lambda_{\alpha+n} \rangle + \pi i Z_{\alpha\alpha}} a_\ell.$$

This tells us that  $\theta$  is completely determined by the coefficients

$$\{a_\ell\}_{0 \leq \ell_\alpha < \delta_\alpha, 1 \leq \alpha \leq n}.$$

Thus, we see that  $H^0(M, \mathcal{L}) \leq \prod_{\alpha=1}^n \delta_\alpha$ . To show  $H^0(M, \mathcal{L}) = \prod_{\alpha=1}^n \delta_\alpha$ , one must demonstrate that any choice of  $a_\ell$  produces a convergent  $\theta$ . Let us rewrite our series  $\theta$  in terms of these  $\{a_\ell\}_{0 \leq \ell_\alpha < \delta_\alpha, 1 \leq \alpha \leq n}$ . We find

$$\theta(z) = \sum_{\ell_0, 0 \leq (\ell_0)_\alpha < \delta_\alpha} \left( \sum_{\ell \in \mathbb{Z}^n} a_{\ell_0 + \Delta_\delta \ell} e^{2\pi i \langle \ell_0 + \Delta_\delta \ell, \Delta_\delta^{-1} z \rangle} \right) = \sum_{\ell_0, 0 \leq (\ell_0)_\alpha < \delta_\alpha} e^{2\pi i \langle \ell_0, \Delta_\delta^{-1} z \rangle} \left( \sum_{\ell \in \mathbb{Z}^n} a_{\ell_0 + \Delta_\delta \ell} e^{2\pi i \langle \ell, z \rangle} \right)$$

If we let  $\theta_{\ell_0} = e^{2\pi i \langle \ell_0, \Delta_\delta^{-1} z \rangle} (\sum_{\ell \in \mathbb{Z}^n} a_{\ell_0 + \Delta_\delta \ell} e^{2\pi i \langle \ell, z \rangle})$  denote the normalized series where  $a_{\ell_0} = 1$ , then we can write

$$\theta(z) = \sum_{\ell_0, 0 \leq (\ell_0)_\alpha < \delta_\alpha} a_{\ell_0} \cdot \theta_{\ell_0}(z).$$

Each coefficient  $a_{\ell_0 + \Delta_\delta \ell}$  equals

$$e^{\pi i \langle \ell, Z \ell \rangle + 2\pi i \langle \Delta_\delta^{-1} \ell_0, Z \ell \rangle} =: b_\ell.$$

Using the fact that  $Z^T = Z$  and  $Ze_\alpha = \lambda_{n+\alpha}$ , one can verify that these  $b_\ell$  satisfy the recursive relation imposed on the coefficients [GH94, Page 320]. Using positive-definiteness of  $\text{Im}(Z)$ , one can produce bounds on  $|b_\ell|$  for  $\ell$  sufficiently large to show that  $\theta$  converges uniformly on compact sets in  $V$  [GH94, Page 320]. Thus, for any choice of  $\{a_\ell\}_{0 \leq \ell_\alpha < \delta_\alpha, 1 \leq \alpha \leq n}$ ,  $\theta(z)$  converges.  $\square$

So in general, the global sections of the line bundle given by the multipliers

$$e_{\lambda_\alpha} \equiv 1 \text{ and } e_{\lambda_{\alpha+n}} = e^{-2\pi i z_\alpha - \pi i Z_{\alpha\alpha}}, 1 \leq \alpha \leq n$$

are given by the  $\prod_{i=1}^n \delta_\alpha$  entire functions

$$\theta(z) = \sum_{\ell_0, 0 \leq (\ell_0)_\alpha < \delta_\alpha} a_{\ell_0} \cdot \theta_{\ell_0}(z)$$

where

$$\theta_{\ell_0}(z) = e^{2\pi i \langle \ell_0, \Delta_\delta^{-1} z \rangle} \cdot \sum_{\ell \in \mathbb{Z}^n} b_\ell e^{2\pi i \langle \ell, z \rangle}, \text{ and } b_\ell = e^{\pi i \langle \ell, Z \ell \rangle + 2\pi i \langle \Delta_\delta^{-1} \ell_0, Z \ell \rangle}.$$



We will now define the Riemann  $\theta$ -function and Theta divisors. Consider the Jacobian  $\mathcal{J}(S) = V/\Lambda$  of a compact Riemann surface of genus  $g$ . There exists integral basis  $\lambda_1, \dots, \lambda_{2n}$  for  $\Lambda$ , and complex basis  $e_j = \lambda_j$  for  $1 \leq j \leq n$ , so that the period matrix of  $\mathcal{J}(S)$  is of the form  $\begin{pmatrix} I & Z \end{pmatrix}$  where  $Z$  is symmetric and  $\text{Im}(Z)$  is positive-definite. In particular,  $\mathcal{J}(S)$  is a principally polarized abelian variety. Letting  $\{z_\alpha\}$  denote complex coordinates on  $V$  dual to  $\{e_\alpha\}$ , and  $\{x_\alpha\}$  denote coordinates on  $V$  dual to  $\{\lambda_\alpha\}$ , there exists an invariant integral Kahler form that may be expressed as

$$\omega = \sum_{\alpha=1}^n dx_\alpha \wedge dx_{\alpha+n}.$$

Consider the line bundle  $\mathcal{L}$  given by the multipliers

$$e_{\lambda_\alpha} \equiv 1 \text{ and } e_{\lambda_{\alpha+n}} = e^{-2\pi i z_\alpha - \pi i Z_{\alpha\alpha}}, \text{ for } 1 \leq \alpha \leq n.$$

Then as previously discussed,  $c_1(\mathcal{L}) = \omega$  and in particular,  $H^0(M, \mathcal{L}) = 1$ . The space of global sections is generated by a particular global section  $\tilde{\theta}$  which is given by a particular entire function  $\theta$  called the *Riemann theta-function*, where

$$\theta = \sum_{\ell \in \mathbb{Z}^n} e^{\pi i \langle \ell, Z \ell \rangle} e^{2\pi i \langle \ell, z \rangle},$$

which satisfies  $\theta(z) = \theta(-z)$  and

$$\theta(z + \lambda_\alpha) = \theta(z) \text{ and } \theta(z + \lambda_{\alpha+n}) = e^{-2\pi i z_\alpha - \pi i Z_{\alpha\alpha}} \cdot \theta(z) \text{ for } 1 \leq \alpha \leq n$$

The effective divisor  $\Theta$  associated to the global section  $\tilde{\theta} \in H^0(M, \mathcal{L})$  given by the Riemann theta function  $\theta$  is called a *Theta divisor*. If  $u \in M$ , then we let  $\Theta_u$  denote the Theta divisor associated to  $\tilde{\theta}_u := \tau_u^* \tilde{\theta} \in H^0(M, \tau_u^* \mathcal{L})$  given by the function  $\tau_u^* \theta = \theta(z - u)$ . Note that  $\Theta_u = \Theta + u$ .

#### 4. SOLVING JACOBI INVERSION

Using Theta divisors, we will solve our Jacobi Inversion problem. We begin by intersecting Theta divisors with the image of our compact Riemann surface  $S$  under the Abel-Jacobi map.

**Theorem 4.1.** *Let  $\mu : S \rightarrow \mathcal{J}(S)$ . Then for  $u \in \mathcal{J}(S)$ , either  $\mu(S) \subset \Theta_u$  or  $\mu(S)$  and  $\Theta_u$  intersect at exactly  $g$  points, counting multiplicity.*

*Proof.* First, suppose that  $\mu(S) \not\subset \Theta$ . To understand how  $\mu(S)$  intersects  $\Theta$ , it suffices to pullback the global section  $\tilde{\theta} \in H^0(\mathcal{J}(S), \mathcal{L})$  to the section  $\mu^* \tilde{\theta} \in H^0(S, \mu^* \mathcal{L})$ . Since  $S$  is a Riemann surface, the effective divisor associated to  $\mu^* \tilde{\theta}$  will be a finite number of points on  $S$ , and these will be the intersection points, counting multiplicity, of  $\mu(S)$  and  $\Theta$ . Now we show that there will always be  $g$  points, counting multiplicity.

If we make the usual identification of  $S$  with a polygon  $\Delta$  of  $4g$  sides, we have a natural lift  $\tilde{\mu} : \Delta \rightarrow V$  such that

$$\tilde{\mu}(z) = \left( \int_b^z \omega_1, \dots, \int_b^z \omega_g \right)$$

fits into the diagram

$$\begin{array}{ccc} \Delta & \xrightarrow{\tilde{\mu}} & V \\ \downarrow & & \downarrow \\ S & \xrightarrow{\mu} & \mathcal{J}(S) \end{array}$$

Then to calculate the points of intersection of  $\mu(S)$  and  $\Theta$ , it suffices to calculate the zeroes of the holomorphic function  $\tilde{\mu}^* \theta$  on  $\Delta$ . We have that the number of zeroes of  $\tilde{\mu}^* \theta$  counting multiplicity is equal to

$$\frac{1}{2\pi i} \int_{\partial \Delta} d \log \theta(\tilde{\mu}(z)).$$

Per the usual technique for integration along the boundary of the polygonal identification of a compact Riemann surface, we first examine integrating only on the sides " $\gamma_\alpha$ " and " $\gamma_\alpha^{-1}$ ", which are identified under  $\Delta \rightarrow S$  with the loop  $\gamma_\alpha$ . First, examine the value of

$$\tilde{\mu}(p') - \tilde{\mu}(p)$$

where  $p \in \gamma_\alpha$  and  $p' \in \gamma_\alpha^{-1}$  are points on the sides of  $\Delta$  identified under  $\Delta \rightarrow S$  with the same point on the loop  $\gamma_\alpha$ . We have that the  $k$ -th coordinate of this quantity is

$$\tilde{\mu}_k(p') - \tilde{\mu}_k(p) = \int_b^{p'} \omega_k - \int_b^p \omega_k = \int_p^{p'} \omega_k = \int_{\gamma_{\alpha+g}} \omega_k = Z_{k\alpha},$$

so  $\tilde{\mu}(p') - \tilde{\mu}(p)$  equals  $\lambda_{\alpha+g}$ . Then we have

$$\theta(\tilde{\mu}(p')) = e^{-2\pi i z_\alpha(\tilde{\mu}(p')) - \pi i Z_{\alpha\alpha}} \theta(\tilde{\mu}(p)) \implies \log(\theta(\tilde{\mu}(p'))) - \log(\theta(\tilde{\mu}(p))) = -2\pi i z_\alpha(\tilde{\mu}(p')) - \pi i Z_{\alpha\alpha}.$$

Thus, we have that

$$\frac{1}{2\pi i} \int_{\gamma_\alpha \cup -\gamma_\alpha^{-1}} d \log \theta(\tilde{\mu}(z)) = \frac{1}{2\pi i} \int_{-\gamma_\alpha^{-1}} d(2\pi i z_\alpha(\tilde{\mu}(z)) + \pi i Z_{\alpha\alpha}) = \int_{\gamma_\alpha} dz_\alpha = 1.$$

Then integrating  $d \log(\theta(\tilde{\mu}(z)))$  along  $\gamma_\alpha$  and  $-\gamma_\alpha^{-1}$  produces the quantity  $g$ .

Now we verify that integrating along the other  $2g$  sides contributes zero. Consider the sides " $\gamma_{\alpha+g}$ " and " $\gamma_{\alpha+g}^{-1}$ ". Let  $p' \in \gamma_{\alpha+g}^{-1}$  and  $p \in \gamma_{\alpha+g}$  be identified. Then we have

$$\tilde{\mu}_k(p') - \tilde{\mu}_k(p) = \int_b^{p'} \omega_k - \int_b^p \omega_k = \int_p^{p'} \omega_k = \int_{-\gamma_\alpha} \omega_k = -\delta_{*\alpha} = -e_\alpha = -\lambda_\alpha,$$

so

$$\theta(\tilde{\mu}(p')) = \theta(\tilde{\mu}(p)) \implies \log \theta(\tilde{\mu}(p')) - \log \theta(\tilde{\mu}(p)) = 0.$$

Thus,

$$\frac{1}{2\pi i} \int_{\gamma_{\alpha+g} \cup -\gamma_{\alpha+g}^{-1}} d \log \theta(\tilde{\mu}(z)) = 0.$$

Thus, we find that

$$\frac{1}{2\pi i} \int_{\partial\Delta} d \log \theta(\tilde{\mu}(z)) = g,$$

which implies our claim that if  $\mu(S) \not\subset \Theta$ , which means that if  $\mu^*\tilde{\theta}$  is not just the zero section, then  $\mu(S)$  and  $\Theta$  intersect at  $g$  points counting multiplicity.

If, however,  $\mu(S) \subset \Theta$ , which means that  $\tilde{\theta}$  pulls back to the zero section of  $\mu^*\mathcal{L}$ , then we can find some  $u \in M$  such that  $\mu(S) \not\subset \Theta_u$  and an analagous argument follows.  $\square$

For  $u \in M$  such that  $\mu(S) \not\subset \Theta_u$ , let  $\mu^*\Theta_u = p_1(u) + \cdots + p_g(u)$  denote the  $g$  points. Amazingly, up to a fixed constant, these points  $p_i(u)$  are the solution to the Jacobi Inversion problem for  $u \in M$ .

**Theorem 4.2.** *There exists a fixed constant  $c \in \mathcal{J}(S)$  such that for all  $u \in M$  such that  $\mu(S) \not\subset \Theta_u$ ,*

$$c + \sum_{i=1}^g \mu(p_i(u)) = u.$$

*Proof.* Just as in the proof of Theorem 4.1, let  $\Delta$  denote the  $4g$ -polygon whose edges glue to  $S$ , and let  $\tilde{\mu} : \Delta \rightarrow V$  denote the lift of  $\mu$ . Fix  $u \in M$  such that  $\mu(S) \not\subset \Theta_u$ . Then  $p_1(u), \dots, p_g(u)$  are the zeroes of  $\theta_u(\tilde{\mu}(z))$ . Fixing the  $k$ -th coordinate for  $1 \leq k \leq g$ , by the residue theorem, we have

$$\sum_{i=1}^g \tilde{\mu}_k(p_i(u)) = \frac{1}{2\pi i} \int_{\partial\Delta} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z)).$$

Just as before, we unpack the integral by examining what happens on sides which glue. First, we consider the sides " $\gamma_\alpha$ " and " $\gamma_\alpha^{-1}$ ". Let  $p' \in \gamma_\alpha^{-1}$  and  $p \in \gamma_\alpha$ . Just as in the proof of Theorem 4.1, we have

$$\tilde{\mu}_k(p') - \tilde{\mu}_k(p) = Z_{k\alpha} \implies \tilde{\mu}(p') - \tilde{\mu}(p) = \lambda_{\alpha+g}.$$

Then

$$\theta_u(\tilde{\mu}(p')) = e^{-2\pi i z_\alpha(\tilde{\mu}(p')) - \pi i Z_{\alpha\alpha} + 2\pi i z_\alpha(u)} \cdot \theta_u(\tilde{\mu}(p)).$$

Then

$$\begin{aligned} \log \theta_u(\tilde{\mu}(p')) - \log \theta_u(\tilde{\mu}(p)) &= -2\pi i z_\alpha(\tilde{\mu}(p')) - \pi i Z_{\alpha\alpha} + 2\pi i z_\alpha(u). \\ \implies d \log \theta_u(\tilde{\mu}(p')) - d \log \theta_u(\tilde{\mu}(p)) &= -2\pi i \omega_\alpha(p'). \end{aligned}$$

Then

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma_\alpha \cup \gamma_\alpha^{-1}} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z)) &= \frac{1}{2\pi i} \int_{\gamma_\alpha} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z)) + \frac{1}{2\pi i} \int_{\gamma_\alpha^{-1}} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z)) \\ &= \frac{1}{2\pi i} \int_{\gamma_\alpha} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z)) + \frac{1}{2\pi i} \int_{\gamma_\alpha} (-\tilde{\mu}_k(z) - Z_{k\alpha}) [d \log \theta_u(\tilde{\mu}(z)) - 2\pi i \omega_\alpha(z)]. \\ &= \int_{\gamma_\alpha} \tilde{\mu}_k(z) \cdot \omega_\alpha(z) - \frac{Z_{k\alpha}}{2\pi i} \int_{\gamma_\alpha} d \log \theta_u(\tilde{\mu}(z)) + Z_{k\alpha}. \end{aligned}$$

The first and third terms are constant and independent of  $u \in M$ , so they may be absorbed into the  $k$ -th coordinate  $c_k$  of the constant  $c$ . We claim that the second term is also constant and can be absorbed into  $c_k$ . Note that if  $x$  and  $x'$  denote the ordered endpoints of  $\gamma_\alpha$ , then  $\tilde{\mu}_k(x') - \tilde{\mu}_k(x) = \int_x^{x'} \omega_k = \int_{\gamma_\alpha} \omega_k = \delta_{k\alpha}$ . Then  $\tilde{\mu}(x') - \tilde{\mu}(x) = e_\alpha$ . Then  $\theta_\lambda$  takes the same value at both  $\tilde{\mu}(x')$  and  $\tilde{\mu}(x)$ . Then  $\theta_u \tilde{\mu}(\gamma_\alpha)$  is a closed loop in  $\mathbb{C}$ , so  $\frac{1}{2\pi i} \int_{\gamma_\alpha} d \log \theta_u(\tilde{\mu}(z)) \in \mathbb{Z}$ , since the integral calculates a winding number. So the integral

$$\frac{1}{2\pi i} \int_{\gamma_\alpha \cup -\gamma_\alpha^{-1}, 1 \leq \alpha \leq n} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z))$$

only contributes to the constant  $c_k$ .

Now let us integrate around the sides " $\lambda_{\alpha+g}$ " and " $\lambda_{\alpha+g}^{-1}$ ". Let  $p \in \lambda_{\alpha+g}$  and  $p' \in \lambda_{\alpha+g}^{-1}$ . Just as in the proof of Theorem 4.1, we have

$$\tilde{\mu}_k(p') - \tilde{\mu}_k(p) = -\delta_{\alpha k} \implies \tilde{\mu}(p') - \tilde{\mu}(p) = -\lambda_\alpha = -e_\alpha \implies \theta_u(\tilde{\mu}(p')) = \theta_u(\tilde{\mu}(p)).$$

We have then that

$$\begin{aligned} \frac{1}{2\pi i} \int_{\gamma_{\alpha+g} \cup -\gamma_{\alpha+g}^{-1}} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z)) &= \frac{1}{2\pi i} \left[ \int_{\gamma_{\alpha+g}} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z)) + \int_{-\gamma_{\alpha+g}^{-1}} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z)) \right] \\ &= \frac{1}{2\pi i} \left[ \int_{\gamma_{\alpha+g}} \tilde{\mu}_k(z) d \log \theta_u(\tilde{\mu}(z)) - \int_{\gamma_{\alpha+g}} (\tilde{\mu}_k(z) - \delta_{\alpha k}) d \log \theta_u(\tilde{\mu}(z)) \right] = \frac{\delta_{\alpha k}}{2\pi i} \int_{\gamma_{\alpha+g}} d \log \theta_u(\tilde{\mu}(z)). \end{aligned}$$

If we let  $x, x'$  denote the ordered endpoints of  $\gamma_{\alpha+g}$ , then  $\tilde{\mu}(p') - \tilde{\mu}(p) = \lambda_{\alpha+g}$  per the proof of Theorem 4.1. This implies that

$$\begin{aligned} \theta_u(\tilde{\mu}(p')) &= e^{-2\pi i z_\alpha(\tilde{\mu}(p')) - \pi i Z_{\alpha\alpha} + 2\pi i z_\alpha(u)} \cdot \theta_u(\tilde{\mu}(p)) \\ \implies \frac{1}{2\pi i} \int_{\gamma_{\alpha+g}} d \log \theta_u(\tilde{\mu}(z)) &= -z_\alpha(\tilde{\mu}(p')) - \frac{\pi i}{2} Z_{\alpha\alpha} + z_\alpha(u) \pmod{\mathbb{Z}} \\ \implies \frac{\delta_{\alpha k}}{2\pi i} \int_{\gamma_{\alpha+g}} d \log \theta_u(\tilde{\mu}(z)) - \delta_{\alpha k} z_\alpha(u) &= -\delta_{\alpha k} z_\alpha(\tilde{\mu}(z)) - \frac{\pi i}{2} Z_{\alpha\alpha} \pmod{\mathbb{Z}}, \end{aligned}$$

and the right hand side are constants that we absorb into  $c_k$ . This shows that for the coordinate  $k$ , there exist constant  $c_k$  such that

$$c_k + \sum_{i=1}^g \tilde{\mu}_k(p_i) = u_k \pmod{\mathbb{Z}},$$

which implies that

$$c + \sum_{i=1}^g \mu(p_i) = u \in \mathcal{J}(S).$$

□

Note for general  $u \in \mathcal{J}(S)$ , we have  $\mu(S) \notin \Theta_u$ . Thus, the Jacobi Inversion problem can be solved for general  $u \in \mathcal{J}(S)$ .

## 5. RIEMANN'S THEOREM AND CONCLUSION

To conclude our exposition, we state what the constant  $c$  in Theorem 4.2 is in terms of the geometry of the Theta divisor  $\Theta$ . This is otherwise known as Riemann's theorem.

Let  $S^{(d)}$  denote the complex manifold which is the quotient manifold of  $S^{\times d}$  by the natural action of the symmetric group  $\Sigma_d$  on  $d$  letters. In other words,  $S^{(d)}$  parameterizes effective divisors of degree  $d$ . Consider the map

$$\mu^{(d)} : S^{(d)} \rightarrow \mathcal{J}(S)$$

given by

$$p_1 + \cdots + p_d \mapsto \left( \sum_{i=1}^d \int_b^{p_i} \omega_1, \dots, \sum_{i=1}^d \int_b^{p_i} \omega_g \right).$$

Suppose  $D = \sum_{i=1}^d p_i \in S^{(d)}$  is an effective divisor such that all the  $p_i$  are distinct. Let  $z_i$  denote local coordinates around each  $p_i$ . Then locally we can write  $\omega_i = \Omega_i(p_j) dz_j$  in terms of the local coordinates  $z_j$  around  $p_j$ . Then the differential of the map  $\mu^{(d)}$  at  $D$  is given by

$$\mathcal{J}(\mu^{(d)}) = \begin{pmatrix} \Omega_1(p_1) & \cdots & \Omega_g(p_1) \\ \vdots & & \vdots \\ \Omega_1(p_d) & \cdots & \Omega_g(p_d) \end{pmatrix}.$$

Note that the complete linear system

$$|\Omega_S| : S \rightarrow \mathbb{P}^{g-1}$$

given by  $\Omega_S$  is always base point free for  $S$  when  $g \geq 2$  by Theorem 6.1. The complete linear system sends

$$p_j \in S \mapsto [\Omega_1(p_j) : \cdots : \Omega_g(p_j)].$$

We see then that  $\mathcal{J}(\Phi)$  has full rank at  $D$  when the images of the  $p_i$  under  $|\Omega_S|$  are linearly independent, which we see is generically the case when  $d \leq g$ . This implies that the image  $W_d = \mu^{(d)}(S^{(d)})$  is an analytic subvariety of dimension  $d$ . Furthermore, the map  $\mu^{(d)}$  is generically one-to-one because the fibers of  $\mu^{(d)}$  are linear spaces. Note that since  $\Theta \subset \mathcal{J}(S)$  is an effective divisor, it is of dimension  $g-1$ . Miraculously,  $c$  is determined by the simple relationship between  $\Theta$  and  $W_{g-1}$ .

**Theorem 5.1** (Riemann's Theorem). *Let  $c$  denote the constant in Theorem 4.2. Then the Theta divisor  $\Theta$  is exactly the translation of  $W_{g-1}$  by  $c$ . In other words,*

$$\Theta = W_{g-1} + c.$$

The proof of Riemann's theorem is described in [GH94, Page 338-340]. In particular, one is also able to analyze exactly when  $\mu(S) \subset \Theta_u$ . Specifically,  $\mu(S) \subset \Theta_c + u$  if and only if  $u = \mu^{(g)}(D)$  for  $D \in S^{(g)}$  such that  $h^0(D) > 0$ . In other words, there fails to be a solution

exactly when the fiber of the Abel-Jacobi map, which is a projective space, has nonzero dimension.

In conclusion, the solution to the Abel-Jacobi Inversion problem is obtained by defining Theta divisors, which are translates of the image  $W_{g-1}$ , and intersecting Theta divisors with the curve  $\mu(S)$ .

## 6. APPENDIX

**Lemma 6.1.** *Let  $X$  be a compact Riemann surface of genus  $g \geq 2$ . The canonical bundle  $\omega_X$  is basepoint free.*

*Proof.* Note by Serre duality we have  $h^1(X, \omega_X) = h^0(X, \mathcal{O}_X) = 1$ , and  $h^1(X, \omega_X(-p)) = h^0(X, \mathcal{O}_X(p))$ . Since  $X$  is genus  $g \geq 2$ , no two points are linearly equivalent (otherwise  $X \cong \mathbb{P}^1$ ). Thus, we must have  $h^0(X, \mathcal{O}_X(p)) = 1$ , since there is a global section with associated effective divisor  $p$ , and any other independent global section would yield an effective divisor of degree 1 i.e. a point  $q$  linearly equivalent to  $p$ , which would be a contradiction. Then we see that  $h^1(X, \omega_X) = h^1(X, \omega_X(-p))$ , and thus by Riemann-Roch, we see that  $h^0(X, \omega_X(-p)) = h^0(X, \omega_X) - 1$ . Thus,  $\omega_X$  is basepoint free.  $\square$

## REFERENCES

- [GH94] Philip Griffiths and Joe Harris. *Principles of Algebraic Geometry*. 1994.
- [Huy05] Daniel Huybrechts. *Complex Geometry*. 2005.